ORF309/MAT380
Poisson Processes I: Definitions

(pp. 97-102)

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Agenda for today

- Recap
- Arrivals as a function of time
- Counting process
- Poisson Processes
- Expectations & Variances
- Examples

Recap

Recap

- The DRV X_i that takes values in $\{0,1\}$, with $P\{X_i = 1\} = p$ and $P\{X_i = 0\} = q$ and $p + q = 1$ is called **Bernoulli RV**

- A **sequence** of n Bernoulli RVs added together is call **Bernoulli Process** and counts the number of successes in n trials

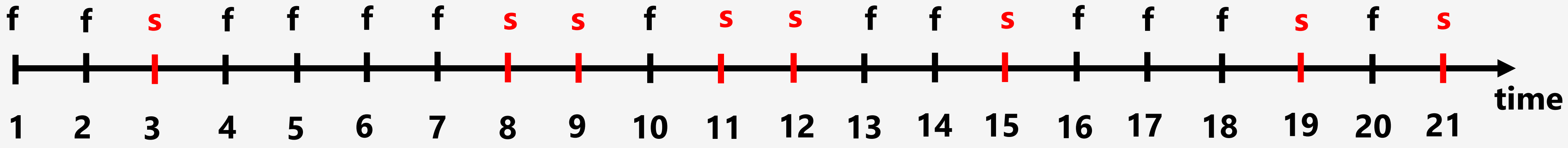
  - $S_n = X_1 + X_2 + \cdots . + X_n$ (S_n is also a RV)

$$\boxed{P\{S_n = k\} = \binom{n}{k} p^k q^{n-k} \quad \text{and} \quad \mathrm{S_n} \sim Binom(n, p)}$$

Time is discrete – Success arrivals occur in predefined known time intervals

Recap

- T_k = "the time of the k-th success"   (in Discrete Time!)
- Its distribution: $P\{T_k = n\}$ for $n \geq 1$



$$P\{T_k = n\} = P\{X_n = 1, S_{n-1} = k\} = P\{X_n = 1\}P\{S_{n-1} = k-1\} = p\binom{n-1}{k-1}p^{k-1}q^{n-k}$$

$$\boxed{T_k \sim NegBinom(k,p)}$$

Recap

- **Special case**: Let's consider: $T_1 = "the\ \ time\ \ unitl\ \ we\ \ get\ \ the\ \ 1st\ \ success"$

$$P\{T_1 = n\} = p\binom{n-1}{1-1}p^{1-1}q^{n-1} = pq^{n-1}$$

- **It is very intuitive:** to get the 1^{st} success in n trials we must get $n-1$ failures

$$Z \sim Geom(p)$$

- **Remark**: $T_i - T_{i-1} \sim Geom(p)$ and $T_i - T_{i-1} \perp\!\!\!\perp T_j - T_{j-1}$

Consecutive increments **Independent increments**

Recap

- **Continuous time Limit (CLT)**

- Defined the DRV: $N =$ **"# of customers that arrive any time within 1 hour (or time unit)"**

- We computed its distribution $P\{N = k\}$ by taking the **limit** of S_n $\quad for\ \ n \to \infty$

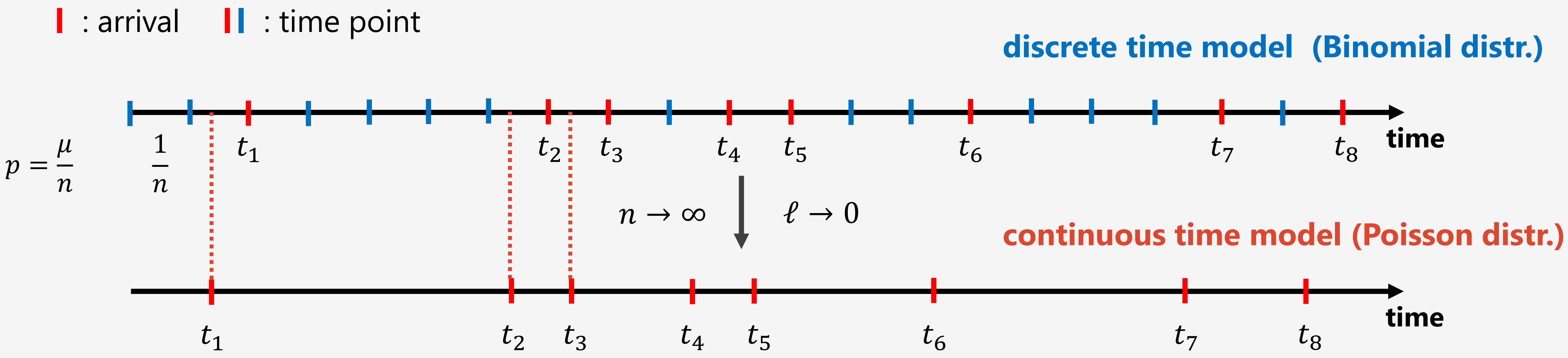
$$P\{N = k\} = \lim_{n\to\infty} P\{S_n = k\} = \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k} = \lim_{n\to\infty} \binom{n}{k} \left(\frac{\boldsymbol{\mu}}{\boldsymbol{n}}\right)^k \left(1 - \frac{\boldsymbol{\mu}}{\boldsymbol{n}}\right)^{n-k} = \frac{\boldsymbol{e^{-\mu}\mu^k}}{\boldsymbol{k}!}$$

❖ **Poisson Distribution**: $N \sim Pois(\mu)$

- We defined the **DRV**: N_t =**"the number of arrivals by time t"** where $t \in [0, \infty)$
(or, in t time units)

❖ $N_t \sim Pois(\mu t)$

Illustration of CTL



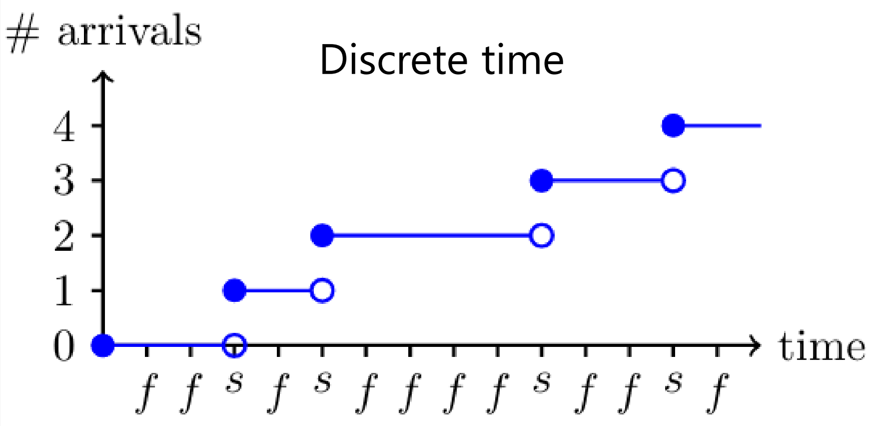
- t_i is the time of the i-th arrival
    - In the discrete time model customers can arrive at specific discrete time points $t_i \in \{1,2,3,\dots\}$
    - In the continuous time model customers can arrive at any time $t_i \in R_+$
- By taking $\boldsymbol{n} \to \infty$ we force $\ell \to \mathbf{0}$ and we convert the discrete time into continuous time

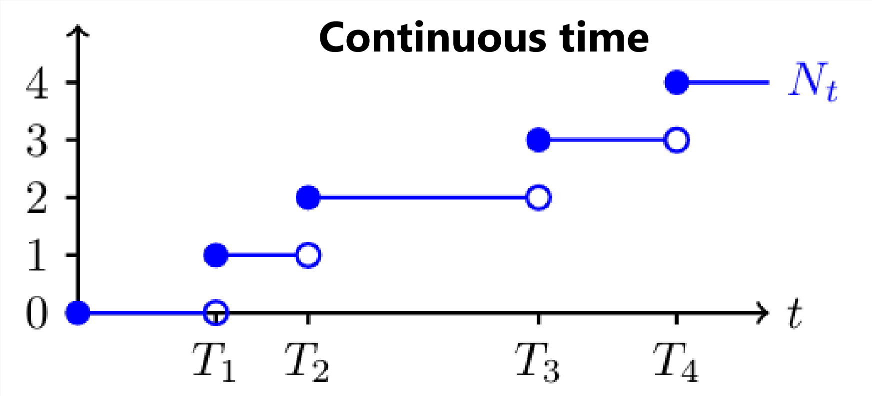
RANDOM PROCESSES

Counting Processes

N_t *as a function of time*

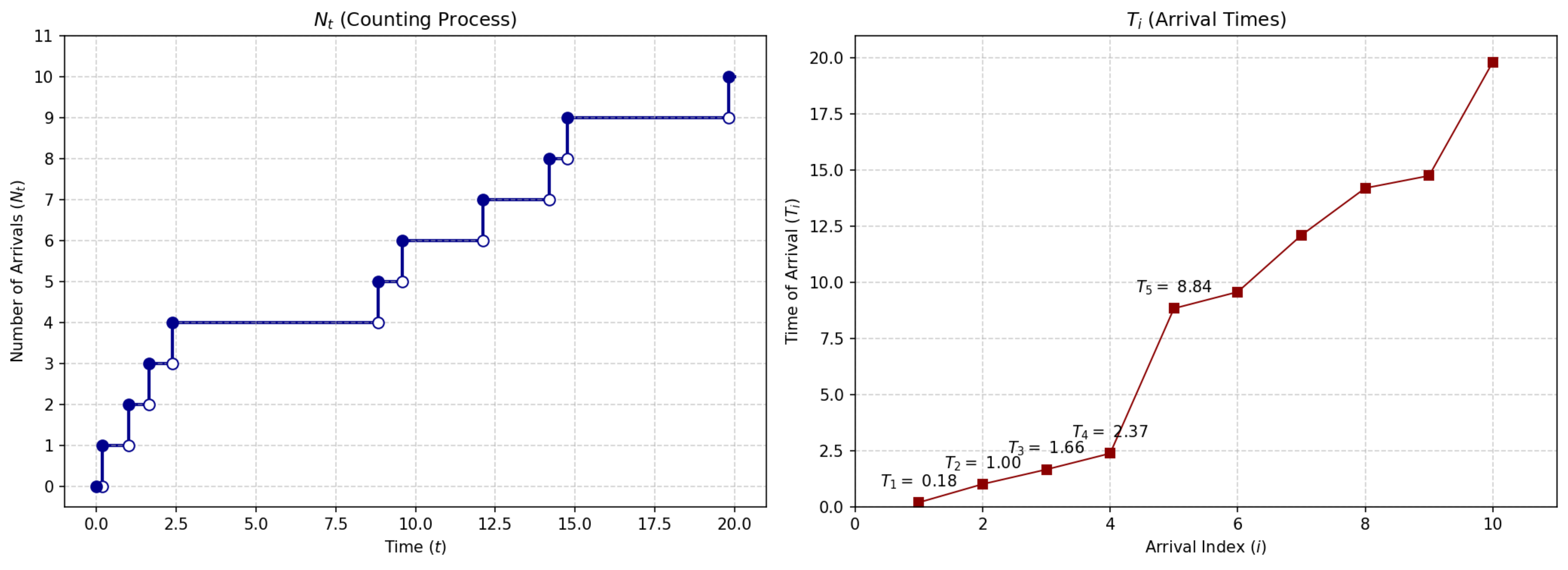
- **In this Lecture:**
  - We will consider N_t as a **function of time**!
    - N_t **increases by 1** every time an arrival occurs
  - T_i is the time of the i-th arrival $(T_i \in \boldsymbol{R}_+)$
  - The **set** $\{N_t\}_{t\geq 0}$ models arrivals in continuous time
    - $\{N_t\}_{t\geq 0}$ is called a **Random Process**
    - N_t **counts the arrivals as they occur** → it is a **Counting Process**





Python simulations

Counting Process simulation of arrivals with $t = 20$ time units



Counting Process

- **Definition**: A random process $\{N_t\}_{t \geq 0}$ is called a **Counting Process** if the following properties hold:

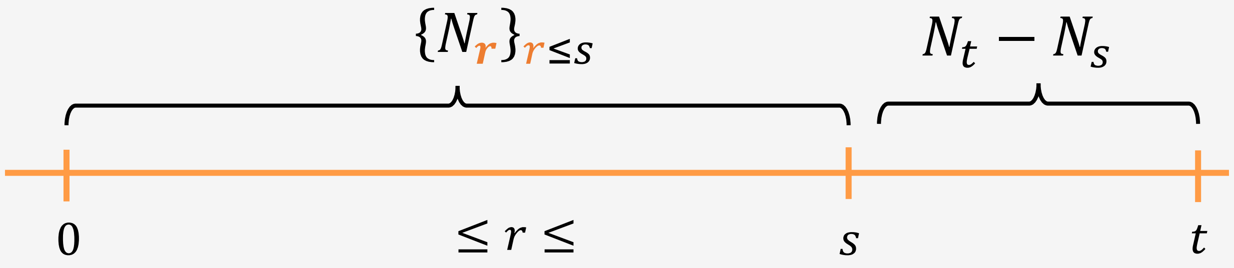
  1. $N_0 = 0$
  2. N_t is a **non-decreasing function** of time t
  3. N_t **increases by 1** every time t an arrivals occurs

- **Remark**: We have **excluded** 2 or more arrivals occurring at the same time, because in continuous time the probability of such arrivals is 0.

- **Note**: The Bernoulli Process is also a **Counting Process.**

Properties of $\boldsymbol{N_t}$ in a Counting Process

- **Property 1**: $N_t \sim Pois(\lambda t),\ t > 0.$
  - We can consider N_t as the **Marginal** of the Random Process, at any fixed t
- **Property 2**: Independent increments
  - $N_t - N_s$="# arrivals in the time interval $(s, t]$" → It is the **increment** of the Counting Process $\{N_t\}_{t\geq 0}$
  - $\{N_r\}_{r\leq s} \perp\!\!\!\perp N_t - N_s$ for all $s < t$



Properties of $\boldsymbol{N_t}$ *in a Counting Process*

- **Property 3**: Stationary increments

- $N_t - N_s \sim N_{\boldsymbol{t-s}} \sim Pois\left(\lambda(\boldsymbol{t-s})\right)$ for $t > s$

- **Remark**: "Stationary" means that the **distribution** of the # of arrivals in time intervals depends on the **length** ($t - s$) of the time intervals, not the times t and s

  - **NOTE**: it is **realistic** for *counting radioactive particles* BUT **NOT realistic** for counting customers arriving at an ice cream store.

- **Remark**: the same property holds for **Bernoulli Process** in discrete time → the number of successes **from trial** $\boldsymbol{m+1}$ **to trial** $\boldsymbol{n}$ and the number of successes **from trial** $\boldsymbol{1}$ **to trial** $\boldsymbol{n-m}$ is the **sum of** $\boldsymbol{n-m}$ **independent Bernoulli variables**

Poisson Processes

Poisson Processes

- We have everything we need to define Poisson processes

- **<u>Definition</u>**: A **counting process** $\{N_t\}_{t \geq 0}$ is called **<u>Poisson process</u>** with rate $\lambda > 0$ if it satisfies the **properties**

   1. $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r \leq s}$ for all $t > s$ (independent increments)

   2. $N_t - N_s \sim Pois\big(\lambda(t-s)\big)$ for $t > s$ (stationary increments) $\Big(N_{t-s} \sim Pois\big(\lambda(t-s)\big)\Big)$

- **Remark**: the rate $\lambda > 0$ can be interpreted as the **<u>average number of arrivals per time unit</u>**.

- **Remark**: the <u>2nd property is not necessary</u> → It can be proved that a counting process $\{N_t\}_{t \geq 0}$ can be a Poisson process if only Property 1 holds.

   - But for this course we will use both properties to characterize a Counting Process as Poisson Process

Poisson Processes

- **Note on new terminology:**
  - $\boldsymbol{N_t}$ is a random variable at any fixed time t that follows $Pois(\lambda t)$, and describes the number of arrivals by that time t
  - $\{\boldsymbol{N_t}\}_{\boldsymbol{t \geq 0}}$ is a collection of random variables with $N_t \sim Pois(\lambda t)$ and is called a Poisson Process
  - The Poisson process $\{\boldsymbol{N_t}\}_{\boldsymbol{t \geq 0}}$ is a **(random) function of time** t, whereas $\boldsymbol{N_t}$ is just a **random variable**!
  - The Poisson Process contains more information, because it contains the **joint distribution** of many Poisson variables. Each $\boldsymbol{N_t}$ is simply the **Marginal** of $\{\boldsymbol{N_t}\}_{\boldsymbol{t \geq 0}}$
  - From now on when we describe a **model of arrivals in continuous time** we will say that the arrivals are modeled according to a **Poisson process** with rate λ (this will tell us everything about the Joint distribution of the arrivals).

Arrivals at different times

- **Example**:

  - $\{N_t\}_{t\geq 0}$ is a Poisson process with rate $\lambda > 0$.

  - **Question**: What is $P\{N_s = i, N_t = j\} =??$ where $s < t$ and $i \leq j$.

Arrivals at different times

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  - $\{N_t\}_{t\geq 0}$ is a Poisson process with rate $\lambda > 0$.

  - <u>**Question**</u>: What is $P\{N_s = i, N_t = j\} =??$ where $s < t$ and $i \leq j$.

$$P\{N_s = i, N_t = j\} = P\{N_s = i, \overset{\text{increment}}{N_t - N_s} = j - i\} \overset{\boxed{\perp\!\!\!\perp}}{=}$$

Arrivals at different times

- **Example**:
  - $\{N_t\}_{t\geq 0}$ is a Poisson process with rate $\lambda > 0$.
  - **<u>Question</u>**: What is $P\{N_s = i, N_t = j\} =??$ where $s < t$ and $i \leq j$.

$$P\{N_s = i, N_t = j\} = P\{N_s = i, \overset{\text{increment}}{N_t - N_s = j - i}\} \overset{\boxed{\perp\!\!\!\perp}}{=}$$

$$= P\{N_s = i\}P\{N_t - N_s = j - i\} = \frac{e^{-\lambda s}(\lambda s)^i}{i!} \cdot \frac{e^{\lambda(t-s)}\big(\lambda(t-s)\big)^{j-i}}{(j-i)!} = \cdots$$

You can simplify it
a little bit, but this is
mostly what you get…

Time of the $\boldsymbol{k}$-th arrival --- $\boldsymbol{T_k}$

Duality: Waiting vs Counting

- **Reminder**: $T_k \sim Gamma(k,\lambda) \rightarrow P\{T_k \le t\} = P\{N_t \ge k\} = 1 - P\{N_t < k\} = 1 - \sum_{i=0}^{k-1} \frac{e^{-\lambda t}(\lambda t)^i}{i!}$
- $T_1, T_2, T_3, \ldots$ are **not** independent CRVs

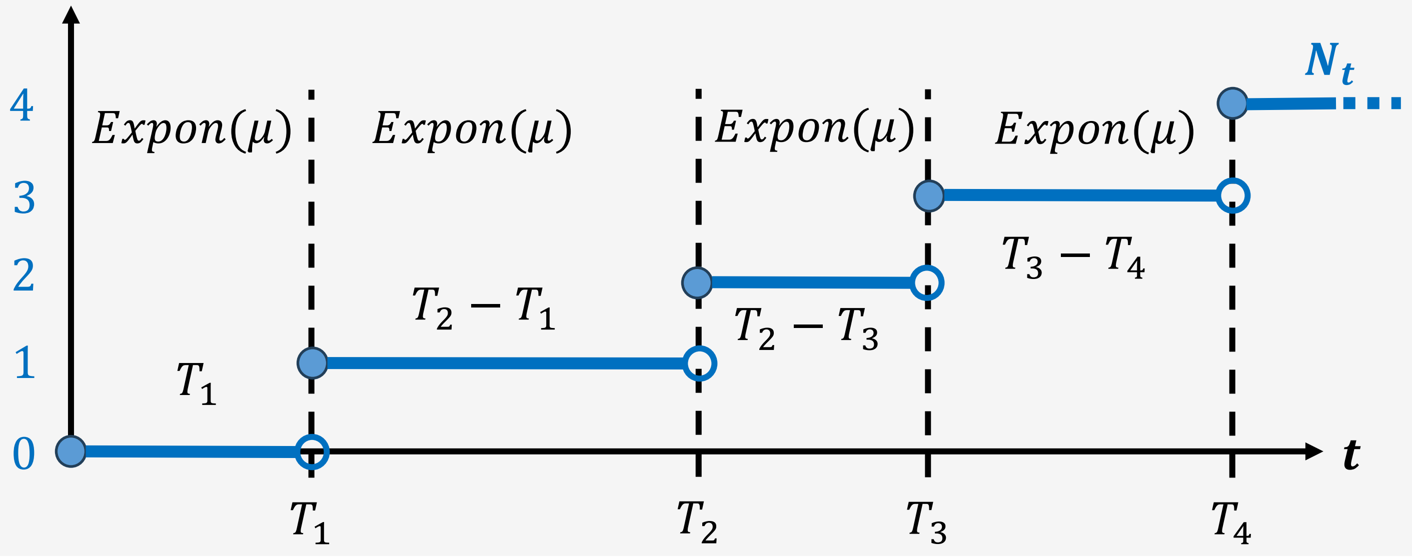
  Example: if $T_1 = 5$ hours, can we have $T_2 = 2$? NO! Hence $T_2 > T_1$
- **Interarrival arrival times**: $T_1, T_2 - T_1, T_3 - T_2, \ldots$ are $\boxed{\perp\!\!\!\perp}$

  $\sim \boldsymbol{Pois}(\boldsymbol{\lambda})$
  - (the 1st arrival will happen in the future) $\{T_1 > t\} = \{N_t = 0\}$ (no arrivals have occurred)
  - $P\{T_1 > t\} = P\{N_t = 0\} = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\mu t} \Rightarrow P\{T_1 \le t\} = 1 - P\{T_1 > t\} = 1 - e^{-\lambda t}$
  - **Hence**: $T_1 \sim Expon(\lambda)$, **and** $T_i - T_{i-1} \sim Expon(\lambda),\ \forall i \rightarrow$ **i.i.d.**

Expectation and Variance of T_k

- Recall the **Telescopic sum**: $T_k = T_k - T_{k-1} + T_{k-1} - T_{k-2} + T_{k-2} - \cdots - T_1 + T_1$
  - $T_k \sim Gamma(k, \mu)$ **and** $T_i - T_{i-1} \sim Expon(\mu)$ **and** $T_i - T_{i-1} \perp\!\!\!\perp T_j - T_{j-1}$
  - Hence T_k can be written as the **sum of** k exponentially distributed random variables

$T_i - T_{i-1}$: waiting period for the i-th arrival



Expectation and Variance of T_k

- **Expectation of T_k:**

$$E[T_k] = E[T_k - T_{k-1} + T_{k-1} - T_{k-2} + T_{k-2} - \cdots + T_2 - T_1 + T_1] =$$

$$= E[T_k - T_{k-1}] + \cdots + E[T_2 - T_1] + E[T_1] = k/\mu$$

- **Variance of T_k:**

$$Var(T_k) = Var(T_k - T_{k-1} + T_{k-1} - T_{k-2} + T_{k-2} - \cdots + T_2 - T_1 + T_1) \overset{\boxed{\perp\!\!\!\perp}}{=}$$

$$Var(T_k - T_{k-1}) + \cdots + Var(T_2 - T_1) + Var(T_1) = k/\mu^2$$

Computations become easier, with no need of the heavy machinery of Calculus

Summary of distributions so far

- **Discrete vs Continuous time**: in a Bernoulli process, arrivals occur only in time intervals whose bounds are predetermined. In Poisson processes, arrivals can occur at any moment in continuous time.

- **The waiting times**: The Geometric distribution models the number of discrete trials until a success in a Bernoulli process. The Exponential distribution is its continuous-time counterpart, and models the actual time spent waiting for an arrival in a Poisson process.

- **Time of the k-th arrival**: The Gamma distribution is the continuous-time counterpart of the Negative Binomial, and they both model the time of the k-th arrival

Summary of distributions so far

- **Resetting the clock ("memorylessness")**: the time increments in both processes are independent of each other. This property can be thought of as resetting the clock to zero, every time an arrival occurs.

- **Independence of interarrival (waiting) times**: by focusing on the times between arrivals $T_i - T_{i-1}$ and noting that they all follow the Exponential distribution, we obtain i.i.d. Exponential random variables.

- **Computational advantage**: using the telescoping sum, we can express T_k as the sum of k independent Exponentially distributed interarrival (waiting) times.

- **Mean & Variance**: this allows us to compute easily: $E[T_k] = k/\mu$ and $Var(T_k) = k/\mu^2$

Summary of our accomplishments so far!!

Arrivals	**Discrete time (Bernoulli Procs)**	**Continuous time (Poisson Procs)**
#arrivals by t	Binomial distribution	Poisson distribution
1^{st} arrival time	Geometric distribution	Exponential distribution
k-th arrival time	Negative Binomial distribution	Gamma distribution
Interarrival times	i.i.d	i.i.d.

We derived all this knowledge by building on the 3 Axioms and Bernoulli random variables!